

Maximizing Orchard Yield and Apiary Sustainability: A Data-Driven Guide for Commercial Pollination

Prepared for commercial beekeeping associations and almond growers

HiMCM 2022 Problem A Team XXXX (plain-language brief)

Dear Association Members and Orchard Partners,

This short guide translates our colony and orchard model into field practice. It is written for people who move bees and sign pollination contracts—not for mathematicians. The numbers below come from our simulation files for a **20-acre almond block** ($\approx 81,000 \text{ m}^2$).

1. How many hives? The golden band

On a 20-acre almond orchard, plan on **1 to 2 hives per acre**, that is **20 to 40 strong colonies** for the whole block. In our optimization, **40 colonies** (2 per acre) gave the best balance of nut set and rental cost at about **85% of full pollination potential** and roughly **\$51,000** net orchard return after a \$200/hive rental. Pushing past about 2.5 hives/acre barely raises set but starts to starve colonies of nectar and pollen (shared bloom), so bees leave the contract weaker.

- **Light bloom / small trees:** start near 1 hive/acre (20 colonies).
- **Full bloom / mature block:** target 1.5–2 hives/acre (30–40 colonies).
- **Avoid:** stacking 3+ hives/acre as a default “insurance” load without checking stores after petal fall.

2. Fan (sector) placement on the block

Scatter colonies so flight paths cover the canopy instead of fighting over the same fence line.

1. Divide the 20 acres into **4–6 sectors** (roughly 3–5 acres each).
2. Place **groups of 4–8 pallets** near sector centers or along alleyways with truck access—not one giant wall of hives on a single edge.
3. Face entrances **southeast** where wind and morning sun allow; keep **8–12 ft** between pallet groups when space allows.
4. Keep a **water source** within a short flight; muddy puddles beat none at all during dry bloom weeks.
5. Record GPS/pallet IDs so weak colonies can be swapped without reshuffling the whole yard.

3. Feeding through bloom

Almond bloom is early and can be cold. Even with orchard nectar, packages often need help.

- **Before move-in:** colonies should be queenright, with sealed brood and enough honey that frames are not light when tipped.
- **During cool bloom (< 55°F flight days):** offer **1:1 sugar syrup** in hive-top feeders and a **pollen substitute patty** if pollen traps/frames look thin.
- **Stop heavy syrup** once consistent flight and orchard nectar flow are obvious—you want bees on flowers, not on feeders.
- **After petal fall:** inspect stores before the next contract; our model treats post-bloom nectar and brood survival as the hive-health score you feel as “spring build-up.”

4. Varroa: the critical control that protects foragers

Our stress tests show that when mites, spray exposure, and cold snaps stack, colonies tip from a safe winter store zone into collapse as **effective forager death** rises. You do not need the equations; you need thresholds you can act on.

- **Monitor:** alcohol wash or sticky board **before** almond staging and again after the contract.
- **Treat on the calendar that matches your region**, but do not enter pollination with high mite loads. As a practical trigger used by many commercial yards: act when washes approach the **high single digits per 300 bees** (follow your association’s current economic threshold—treat *before* the contract if you are already close).
- **Rotate modes of action;** never rely on a single chemical year after year.
- **Coordinate with the grower** on fungicide/insecticide timing. Night sprays and avoiding open-bloom insecticides protect the foragers your rent is paying for.

5. Shared checklist (grower + beekeeper)

1. Contract states hive count in the **20–40** band for 20 acres, colony strength standard, and payment timing.
2. Map shows **sector/fan placement** and water.
3. Spray schedule shared **72 hours** ahead when possible.
4. Post-bloom: store check + mite check before the next move.

Healthy bees and full almond sets are the same project. Stock the golden band, place colonies so they cover the canopy, feed only when weather steals flight, and keep Varroa from quietly raising forager death during the most valuable weeks of the year.

Respectfully,

Team XXXX

HiMCM 2022 Problem A Modeling Group